

Unit 11. Linear Models and Estimation

1. Means, covariance, and a fixed design

For a random column $Y \in \mathbb{R}^n$ with finite second moments, define

$$\mu = EY, \quad \text{Cov}(Y) = E[(Y - \mu)(Y - \mu)^\top].$$

Expectation of a vector or matrix is taken entry by entry. A covariance matrix is symmetric and positive semidefinite, since $a^\top \text{Cov}(Y)a = E[(a^\top(Y - \mu))^2] \geq 0$. For deterministic matrices A, B and a deterministic column c ,

$$E(AY + c) = A\mu + c, \quad \text{Cov}(AY) = A \text{Cov}(Y) A^\top, \quad \text{Cov}(AY, BY) = A \text{Cov}(Y) B^\top.$$

For example, the last identity follows by substituting $AY - E(AY) = A(Y - \mu)$ and $BY - E(BY) = B(Y - \mu)$ in the definition of cross-covariance and moving the fixed matrices outside the expectation.

Proposition 11.1 (Expected quadratic form). For a fixed $n \times n$ matrix A ,

$$E(Y^\top AY) = \text{tr}(A \text{Cov}(Y)) + \mu^\top A\mu.$$

Proof. Write $Z = Y - \mu$, so $EZ = 0$. Expanding $Y = \mu + Z$ makes the two cross terms have expectation zero. The remaining random term satisfies

$$E(Z^\top AZ) = \sum_{i,j} a_{ij} E(Z_i Z_j) = \sum_{i,j} a_{ij} \text{Cov}(Y)_{ji} = \text{tr}(A \text{Cov}(Y)).$$

Symmetry of A is not needed for the identity. □

The **linear model** is

$$Y = X\beta + \varepsilon, \quad E\varepsilon = 0, \quad \text{Cov}(\varepsilon) = \sigma^2 I_n, \quad \sigma^2 > 0.$$

Here $X \in \mathbb{R}^{n \times p}$ is a fixed known design matrix and $\beta \in \mathbb{R}^p$ is unknown. The response Y and error ε are random; an observed response is denoted by y . Throughout Sections 1–5 assume X has full column rank. For residual variance estimation require $p < n$. An intercept is present precisely when the constant column $\mathbf{1} = (1, \dots, 1)^\top$ belongs to $\text{col}(X)$. The model's mean vectors range over this column space. No normality assumption is needed for the moment and optimality results below.

2. Fitted vectors, residuals, and variance

Put

$$K = X^\top X, \quad L = K^{-1} X^\top, \quad H = XL = XK^{-1} X^\top, \quad M = I_n - H.$$

The matrix K is positive definite because $a^\top Ka = \|Xa\|^2 > 0$ when $a \neq 0$. The least-squares results from Unit 10 give

$$\hat{\beta} = LY, \quad \hat{Y} = HY, \quad e = MY.$$

The matrices H and M are the orthogonal projectors onto $\text{col}(X)$ and $\ker X^\top$, respectively. In particular

$$H^\top = H = H^2, \quad M^\top = M = M^2, \quad HX = X, \quad MX = 0, \quad HM = MH = 0.$$

For instance, $H^2 = XK^{-1}KK^{-1}X^\top = H$; symmetry follows from symmetry of K^{-1} . Also $\text{col}(H) = \text{col}(X)$ because H takes values there and fixes every Xa . Hence

$$\text{rank } H = \text{tr } H = p, \quad \text{rank } M = \text{tr } M = n - p.$$

The trace equality follows either from eigenvalues 0, 1 of a symmetric idempotent matrix or from $\text{tr } H = \text{tr}(K^{-1}X^\top X) = p$.

Theorem 11.2 (First and second moments of the fit). Under the linear-model assumptions,

$$\begin{aligned} \mathbb{E} \hat{\beta} &= \beta, & \text{Cov}(\hat{\beta}) &= \sigma^2 K^{-1}, \\ \mathbb{E} \hat{Y} &= X\beta, & \text{Cov}(\hat{Y}) &= \sigma^2 H, \\ \mathbb{E} e &= 0, & \text{Cov}(e) &= \sigma^2 M, & \text{Cov}(\hat{\beta}, e) &= 0. \end{aligned}$$

If $p < n$, $s^2 = \|e\|^2 / (n - p)$ is unbiased for σ^2 .

Proof. Since $LX = I_p$, $\hat{\beta} = \beta + L\varepsilon$, giving its mean and covariance $\sigma^2 LL^\top = \sigma^2 K^{-1}KK^{-1} = \sigma^2 K^{-1}$. Likewise $\hat{Y} = X\beta + H\varepsilon$ and $e = M\varepsilon$. Idempotence yields $\text{Cov}(\hat{Y}) = \sigma^2 H$ and $\text{Cov}(e) = \sigma^2 M$. Their cross-covariance with the coefficients is $\sigma^2 LM = 0$, because $X^\top M = 0$. Finally Proposition 11.1 gives $\mathbb{E} \|e\|^2 = \mathbb{E}(\varepsilon^\top M\varepsilon) = \sigma^2 \text{tr } M = \sigma^2(n - p)$. Divide by the positive number $n - p$. $\square$

The model errors and residuals are different vectors: $\varepsilon = Y - X\beta$, whereas $e = Y - X\hat{\beta} = M\varepsilon$. The residuals obey exact constraints $X^\top e = 0$. Their variances are $\sigma^2(1 - h_{ii})$ and their off-diagonal covariances are $-\sigma^2 h_{ij}$. Zero cross-covariance of estimates and residuals does not by itself establish independence.

Example 11.1 (A fit, its geometry, and its uncertainty). Take

$$t = (-3, -1, 1, 3)^\top, \quad X = (\mathbf{1} \ t), \quad y = (1, 2, 2, 5)^\top.$$

Here $\mathbf{1}^\top t = 0$ and $t^\top t = 20$, so

$$K = \begin{pmatrix} 4 & 0 \\ 0 & 20 \end{pmatrix}, \quad X^\top y = \begin{pmatrix} 10 \\ 12 \end{pmatrix}, \quad \hat{\beta} = \begin{pmatrix} 5/2 \\ 3/5 \end{pmatrix}.$$

The fitted and residual columns are

$$\hat{y} = \frac{1}{10}(7, 19, 31, 43)^\top, \quad e = \frac{1}{10}(3, 1, -11, 7)^\top.$$

Both $\mathbf{1}^\top e = 0$ and $t^\top e = 0$ hold by direct substitution. The residual sum of squares is $\text{SSE} = 180/100 = 9/5$, and $s^2 = (9/5)/(4 - 2) = 9/10$. The covariance of the coefficient estimator is $\sigma^2 \text{diag}(1/4, 1/20)$; replacing σ^2 by s^2 estimates it by $\text{diag}(9/40, 9/200)$.

The projector has entries $h_{ij} = 1/4 + t_i t_j / 20$, so its diagonal is $(7/10, 3/10, 3/10, 7/10)$. Thus the endpoint residuals have variance $3\sigma^2/10$, and the middle residuals have variance $7\sigma^2/10$. These are sampling variances of random residuals; they are not the squares of the observed residual entries.

Proposition 11.3 (Leverage). For every i , $0 \leq h_{ii} \leq 1$, and $\sum_i h_{ii} = p$. If an intercept is present, $h_{ii} \geq 1/n$.

Proof. Let d_i be the i th coordinate vector of $\mathbb{R}^n$. Since H is a projector, $h_{ii} = d_i^\top H d_i = \|H d_i\|^2 \geq 0$. The orthogonal decomposition $d_i = H d_i + M d_i$ gives $1 = h_{ii} + \|M d_i\|^2$, proving the upper bound. Summing the diagonal gives $\text{tr } H = p$. If $\mathbf{1} \in \text{col}(X)$, write $J = \mathbf{1}\mathbf{1}^\top / n$. Then $HJ = JH = J$, and $H - J$ is symmetric idempotent. Its diagonal is nonnegative, so $h_{ii} - 1/n \geq 0$. $\square$

3. The Gauss–Markov theorem

A **linear estimator** of β has the form BY for a fixed $p \times n$ matrix B . It is **unbiased** for every β exactly when $BX = I_p$. Indeed $\mathbb{E}(BY) = BX\beta$, and equality to β for all β is equivalent to equality of those matrices. For symmetric matrices, $C \succeq D$ means $C - D$ is positive semidefinite. This compares the variances of *all* scalar linear combinations: $a^\top C a \geq a^\top D a$ for every a .

Theorem 11.4 (Gauss–Markov). Among all linear unbiased estimators of β , $\hat{\beta} = LY$ has the smallest covariance matrix in the positive-semidefinite order. More precisely, for any B with $BX = I_p$,

$$\text{Cov}(BY) - \text{Cov}(LY) = \sigma^2(B - L)(B - L)^\top \succeq 0.$$

Equality of the covariance matrices holds exactly when $B = L$.

Proof. Set $D = B - L$. Since $BX = LX = I_p$, $DX = 0$. Consequently $LD^\top = K^{-1}X^\top D^\top = 0$ and $DL^\top = 0$. Expansion gives $BB^\top = LL^\top + DD^\top$. Multiplying by σ^2 proves the covariance identity; for every a , $a^\top DD^\top a = \|D^\top a\|^2 \geq 0$. If the covariance difference is zero, its trace is $\sigma^2 \sum_{i,j} d_{ij}^2 = 0$, forcing $D = 0$. Conversely $D = 0$ plainly gives equality. $\square$

For a single target $c^\top \beta$, equality of its variance means $D^\top c = 0$; it need not force every row of D to vanish. Unbiasedness is a condition for all parameter values. Choosing a matrix using the observed responses would no longer give an estimator linear in Y in the sense used by this theorem.

Example 11.2 (The excess variance is computable). Continue Example 11.1. Put $d = (1, -1, -1, 1)^\top$. Then $d^\top \mathbf{1} = d^\top t = 0$. For any fixed real a , define

$$\tilde{\beta}_0 = \hat{\beta}_0 + a d^\top Y, \quad \tilde{\beta}_1 = \hat{\beta}_1.$$

The extra term has mean $a d^\top X \beta = 0$, so both estimators remain unbiased. Since $\|d\|^2 = 4$, the covariance difference is $\sigma^2 \text{diag}(4a^2, 0)$. The slope variance is unchanged; the intercept variance increases strictly unless $a = 0$. With the observed data, $d^\top y = 2$.

4. Removing nuisance columns and comparing models

Partition a full-column-rank design as $X = (N \ Z)$, where N contains the nuisance columns. Write the coefficient column as $(\alpha^\top, \gamma^\top)^\top$. Let $H_N = N(N^\top N)^{-1}N^\top$ and $M_N = I - H_N$.

Theorem 11.5 (Partitioned regression). The matrix $Z^\top M_N Z$ is positive definite, and

$$\hat{\gamma} = (Z^\top M_N Z)^{-1} Z^\top M_N Y, \quad \hat{\alpha} = (N^\top N)^{-1} N^\top (Y - Z \hat{\gamma}).$$

The full-model residual equals $M_N Y - M_N Z \hat{\gamma}$.

Proof. If $M_N Z v = 0$, then $Z v \in \text{col}(N)$, say $Z v = N u$. The relation $N(-u) + Z v = 0$ and full column rank of $(N \ Z)$ give $v = 0$. Thus $v^\top Z^\top M_N Z v = \|M_N Z v\|^2 > 0$ for $v \neq 0$.

For a fixed γ , project $Y - Z \gamma$ onto $\text{col}(N)$. The unique minimizing α is $(N^\top N)^{-1} N^\top (Y - Z \gamma)$, and the remaining residual is $M_N (Y - Z \gamma)$. Minimizing its squared norm is now an ordinary least-squares problem with response $M_N Y$ and design $M_N Z$, of full column rank as just proved. Its normal equations give the displayed $\hat{\gamma}$. Substitution yields $\hat{\alpha}$ and the residual. $\square$

Example 11.3 (Centering a simple regression). Let $N = \mathbf{1}$ and let $Z = z$ be a nonconstant observation column. Then $M_N z = z - \bar{z} \mathbf{1}$ and $M_N y = y - \bar{y} \mathbf{1}$. Partitioned regression gives

$$\hat{\gamma} = \frac{\sum_i (z_i - \bar{z})(y_i - \bar{y})}{\sum_i (z_i - \bar{z})^2}, \quad \hat{\alpha} = \bar{y} - \hat{\gamma} \bar{z}.$$

Centering removes the intercept before estimating the slope. In Example 11.1, $z = t$ has mean zero, the numerator is 12 and the denominator is 20, recovering $\hat{\gamma} = 3/5$ and $\hat{\alpha} = 5/2$. If all z_i are equal, the denominator is zero: the intercept and regressor columns are dependent, so separate intercept and slope coefficients are not identifiable.

If $\mathcal{S}_0 \subseteq \mathcal{S}_1$ are two model spaces, their orthogonal projectors satisfy $H_1 H_0 = H_0 H_1 = H_0$. The difference $H_1 - H_0$ is the orthogonal projector onto $\mathcal{S}_1 \cap \mathcal{S}_0^\perp$, since its square equals itself and it fixes exactly that subspace. Therefore

$$\|(I - H_0)y\|^2 = \|(H_1 - H_0)y\|^2 + \|(I - H_1)y\|^2.$$

Adding columns cannot increase the least-squares residual sum of squares. It decreases it strictly exactly when $(H_1 - H_0)y \neq 0$.

When $\mathbf{1} \in \text{col}(X)$, use $H_0 = J = \mathbf{1}\mathbf{1}^\top/n$ to obtain

$$\underbrace{\|y - \bar{y}\mathbf{1}\|^2}_{\text{SST}} = \underbrace{\|\hat{y} - \bar{y}\mathbf{1}\|^2}_{\text{SSR}} + \underbrace{\|y - \hat{y}\|^2}_{\text{SSE}}.$$

For $\text{SST} > 0$, $R^2 = 1 - \text{SSE}/\text{SST}$ lies in $[0, 1]$. For the running fit, $\text{SST} = 9$, $\text{SSR} = 36/5$ and $\text{SSE} = 9/5$, giving $R^2 = 4/5$. Without a constant column, this centered orthogonal decomposition need not hold; the always-valid decomposition is $\|y\|^2 = \|Hy\|^2 + \|My\|^2$.

5. Generalized least squares

Now suppose $\text{Cov}(\varepsilon) = \sigma^2\Omega$, where Ω is known, symmetric and positive definite. The means are unchanged. Let $W = \Omega^{-1/2}$, the symmetric inverse square root supplied by the spectral theorem. Then

$$\tilde{Y} = WY, \quad \tilde{X} = WX, \quad \tilde{\varepsilon} = W\varepsilon, \quad \text{Cov}(\tilde{\varepsilon}) = \sigma^2 I.$$

This invertible change of observation coordinates is called whitening.

Theorem 11.6 (Generalized least squares and Aitken's theorem). Let $K_\Omega = X^\top \Omega^{-1} X$. The unique minimizer of $(Y - Xb)^\top \Omega^{-1} (Y - Xb)$ is

$$\hat{\beta}_\Omega = L_\Omega Y, \quad L_\Omega = K_\Omega^{-1} X^\top \Omega^{-1}.$$

It is unbiased, with covariance $\sigma^2 K_\Omega^{-1}$, and is best among all linear unbiased estimators in the covariance order.

Proof. The objective is $\|WY - WXb\|^2$ and WX has full column rank. Ordinary least squares on the whitened model gives the formula, its unbiasedness and covariance. Alternatively, $L_\Omega X = I$ and $L_\Omega \Omega L_\Omega^\top = K_\Omega^{-1}$ verify these directly. For any B with $BX = I$, set $D = B - L_\Omega$. Then $DX = 0$ and $L_\Omega \Omega D^\top = K_\Omega^{-1} X^\top D^\top = 0$. Hence

$$\text{Cov}(BY) - \text{Cov}(L_\Omega Y) = \sigma^2 D \Omega D^\top \succeq 0.$$

Because Ω is positive definite, equality holds exactly when $D = 0$. □

The original-coordinate fitted matrix $H_\Omega = X L_\Omega$ is idempotent. It is generally not symmetric, but $H_\Omega^\top \Omega^{-1} = \Omega^{-1} H_\Omega$. The residual satisfies $X^\top \Omega^{-1} (Y - X \hat{\beta}_\Omega) = 0$: orthogonality is measured by the weighted inner product $\langle u, v \rangle_\Omega = u^\top \Omega^{-1} v$. In contrast, the ordinary estimator LY is still unbiased, but its actual covariance is the sandwich matrix $\sigma^2 K^{-1} X^\top \Omega X K^{-1}$.

Example 11.4 (Two observations with unequal precision). For $X = (1, 1)^\top$ and $\Omega = \text{diag}(1, 4)$,

$$\hat{\beta}_\Omega = \frac{4}{5} Y_1 + \frac{1}{5} Y_2, \quad \text{Var}(\hat{\beta}_\Omega) = \frac{4}{5} \sigma^2.$$

Ordinary least squares gives $(Y_1 + Y_2)/2$ with variance $5\sigma^2/4$. For $y = (1, 3)^\top$, the estimates are $7/5$ and 2 , respectively. The weighted residual is $(-2/5, 8/5)^\top$, and its weighted sum is $-2/5 + (1/4)(8/5) = 0$.

The unweighted sum is $6/5$. Here $H_\Omega = \begin{pmatrix} 4/5 & 1/5 \\ 4/5 & 1/5 \end{pmatrix}$ is visibly nonsymmetric, while it still fixes the constant columns and satisfies $H_\Omega^2 = H_\Omega$.

6. Constraints and nonidentifiable coefficients

The following extensions use the same orthogonal-decomposition argument. They also supply the geometric tools for the more substantial exercises. Let R be $q \times p$ of full row rank and impose $R\beta = r$. The full-rank weighted objective splits as

$$Q(b) = Q(\hat{\beta}_\Omega) + (b - \hat{\beta}_\Omega)^\top K_\Omega (b - \hat{\beta}_\Omega),$$

because the weighted normal equations make the cross term zero. Put $C = RK_\Omega^{-1}R^\top$; it is positive definite since $R^\top a \neq 0$ when $a \neq 0$.

Proposition 11.7 (Constrained least squares). The unique constrained minimizer is

$$\hat{\beta}_R = \hat{\beta}_\Omega - K_\Omega^{-1}R^\top C^{-1}(R\hat{\beta}_\Omega - r).$$

Its increase in the objective is $(R\hat{\beta}_\Omega - r)^\top C^{-1}(R\hat{\beta}_\Omega - r)$.

Proof. The formula gives $R\hat{\beta}_R = r$, so it is feasible. If another feasible vector is $\hat{\beta}_R + v$, then $Rv = 0$. The cross term in its K_Ω -distance from $\hat{\beta}_\Omega$ is zero because

$$v^\top K_\Omega(\hat{\beta}_R - \hat{\beta}_\Omega) = -v^\top R^\top C^{-1}(R\hat{\beta}_\Omega - r) = 0.$$

The additional objective is thus $v^\top K_\Omega v$, strictly positive unless $v = 0$. Substitution of the displayed displacement gives the claimed increase, using $RK_\Omega^{-1}R^\top = C$. $\square$

For the running fit, imposing slope zero gives $R = (0 \ 1)$, $r = 0$, $C = 1/20$. The constrained intercept remains $5/2$ and the increase is $(3/5)^2/(1/20) = 36/5$. Its SSE is $9/5 + 36/5 = 9$, the total centered sum of squares, agreeing with the constant-only model.

If X has rank $r < p$, the mean $X\beta$ is identifiable, but individual coefficients may not be. Parameter columns β and $\beta + v$ with $v \in \ker X$ describe the same mean. A scalar target $c^\top \beta$ is **estimable** if some linear estimator $a^\top Y$ is unbiased for it for every β .

Theorem 11.8 (Estimable targets). The target $c^\top \beta$ is estimable exactly when $c \in \text{col}(X^\top) = (\ker X)^\perp$. Under spherical covariance, $c^\top X^+ Y$ is its best linear unbiased estimator. The fitted vector is $XX^+ Y$, and $\|(I - XX^+)Y\|^2/(n - r)$ is unbiased for σ^2 when $r < n$.

Proof. Unbiasedness of $a^\top Y$ means $a^\top X\beta = c^\top \beta$ for every β , or $X^\top a = c$. This proves the first equivalence; the orthogonal-complement equality is the fundamental row/kernel relation. For estimable c , $c^\top X^+ X = c^\top$, since $X^+ X$ projects onto the row space. Thus $a_0 = (X^+)^\top c$ satisfies $X^\top a_0 = c$. Any other solution is $a = a_0 + d$ with $X^\top d = 0$. The SVD shows $a_0 \in \text{col}(X)$, so $a_0^\top d = 0$. Consequently $\text{Var}(a^\top Y) = \sigma^2 \|a_0\|^2 + \sigma^2 \|d\|^2$, proving optimality. The last two claims follow from the projector $H = XX^+$ of rank r and the earlier moment proof, with p replaced by r . $\square$

For $X = (\mathbf{1} \ t \ 2t)$ with the centered nonzero column t from Example 11.1, the mean is $\beta_0 \mathbf{1} + (\beta_1 + 2\beta_2)t$. The estimable targets are exactly those with coefficient vector $(c_0, c_1, 2c_1)^\top$. The separate slopes are not estimable. All least-squares coefficient columns satisfy $b_0 = 5/2$, $b_1 + 2b_2 = 3/5$. Their minimum-norm choice is $(5/2, 3/25, 6/25)^\top$, obtained by projecting (b_1, b_2) onto the direction $(1, 2)$. Fitted values and SSE are unchanged by this redundant parameterization.

Exercises and solutions

Projection, moments, and ordinary least squares

Unless stated otherwise in this section, $X \in \mathbb{R}^{n \times p}$ is fixed and has full column rank, and $Y = X\beta + \varepsilon$ with $E\varepsilon = 0$ and $\text{Cov}(\varepsilon) = \sigma^2 I_n$, $\sigma^2 > 0$. Residual variance estimation requires $n > p$.

Exercise 11.1 (The residual projector). Let $X \in \mathbb{R}^{n \times p}$ have rank p , and set $H = X(X^\top X)^{-1}X^\top$, $M = I - H$. Prove that H, M are symmetric idempotent, $HX = X$, $MX = 0$, $HM = 0$, and that their ranks and traces are p and $n - p$.

Solution. Put $K = X^\top X$. Since K is symmetric positive definite, K^{-1} is symmetric. Hence $H^\top = H$, and $H^2 = XK^{-1}KK^{-1}X^\top = H$. Also $HX = XK^{-1}K = X$. It follows that $M^\top = M$, $M^2 = I - 2H + H^2 = M$, $MX = 0$, and $HM = H - H^2 = 0$. The range of H is exactly $\text{col}(X)$: its outputs lie there and it fixes every column of X . Thus its rank is p . The complementary range of M is $\ker X^\top$, of dimension $n - p$. An idempotent symmetric matrix has eigenvalues 0 and 1, because $Av = \lambda v$ gives $\lambda^2 v = \lambda v$. Its trace and rank both count its eigenvalues 1, proving the trace claims.

Exercise 11.2 (Characterizing a projector by its action). Let $A = A^\top = A^2$ and let $B \in \mathbb{R}^{n \times m}$ have full column rank. Prove (a) if $AB = B$ and $\text{rank } A = m$, then $A = B(B^\top B)^{-1}B^\top$; (b) if $AB = 0$ and $\text{rank } A + m = n$, then $A = I - B(B^\top B)^{-1}B^\top$.

Solution. (a) The equality $AB = B$ puts $\text{col}(B)$ inside $\text{col}(A)$. Their dimensions are both m , so the subspaces coincide. A symmetric idempotent matrix is the orthogonal projector onto its range: it fixes its range and kills its orthogonal complement. Both A and $B(B^\top B)^{-1}B^\top$ have this action, hence are equal. (b) The matrix $I - A$ is symmetric idempotent, fixes B , and has $\text{rank } I - A = m$. Apply (a) to $I - A$ and rearrange.

Exercise 11.3 (The unique least-squares minimizer). For full-column-rank X and arbitrary y , find the unique minimizer of $\|y - Xb\|^2$ and prove global minimality.

Solution. Set $b_0 = (X^\top X)^{-1}X^\top y$ and $r = y - Xb_0$. Then $X^\top r = 0$. For any b ,

$$\|y - Xb\|^2 = \|r - X(b - b_0)\|^2 = \|r\|^2 + (b - b_0)^\top X^\top X(b - b_0).$$

The second term is positive unless $b = b_0$, since X has trivial kernel. Thus b_0 is the unique global minimizer. This proof checks global minimality directly, without a second-derivative assumption.

Exercise 11.4 (Unbiased coefficients and covariance). Under $Y = X\beta + \varepsilon$, $E\varepsilon = 0$ and $\text{Cov}(\varepsilon) = \sigma^2 I$, prove that the least-squares coefficient estimator is unbiased and find its covariance.

Solution. Let $L = (X^\top X)^{-1}X^\top$. Since $LX = I$, $\hat{\beta} = \beta + L\varepsilon$. Thus $E\hat{\beta} = \beta + LE\varepsilon = \beta$, and

$$\text{Cov}(\hat{\beta}) = L(\sigma^2 I)L^\top = \sigma^2(X^\top X)^{-1}X^\top X(X^\top X)^{-1} = \sigma^2(X^\top X)^{-1}.$$

Full column rank is needed even for defining this formula.

Exercise 11.5 (Unbiased residual variance). For $p < n$, prove that $s^2 = Y^\top MY / (n - p)$ is unbiased for σ^2 .

Solution. Since $MX = 0$, $MY = M\varepsilon$, and $Y^\top MY = \varepsilon^\top M\varepsilon$. Expanding by entries gives $E(\varepsilon^\top M\varepsilon) = \sum_{i,j} m_{ij} E(\varepsilon_i \varepsilon_j) = \sigma^2 \sum_i m_{ii} = \sigma^2(n - p)$. Division proves the result. If $p = n$, then $M = 0$, so SSE is always zero and this estimator cannot be defined by dividing by residual degrees of freedom.

Exercise 11.6 (Leverages and residual variances). Prove $0 \leq h_{ii} \leq 1$, $\sum_i h_{ii} = p$, and $\text{Var}(e_i) = \sigma^2(1 - h_{ii})$ for $e = MY$.

Solution. For the coordinate column d_i , $h_{ii} = \|Hd_i\|^2 \geq 0$. The orthogonal sum $d_i = Hd_i + Md_i$ gives $1 = h_{ii} + \|Md_i\|^2$, proving the upper bound. The sum equals $\text{tr } H = p$. Finally $e = M\varepsilon$ and $\text{Cov}(e) = \sigma^2 MM^\top = \sigma^2 M$, whose i th diagonal is the claimed variance. No independence of the residual coordinates is implied.

Exercise 11.7 (A numerical fit and its covariance). For $X = (\mathbf{1} \ t)$, $t = (-3, -1, 1, 3)^\top$, and $y = (1, 2, 2, 5)^\top$, compute the least-squares coefficients, fitted values, residuals, SSE, s^2 , and the estimated covariance of the coefficient estimator. Verify the two residual orthogonality constraints.

Solution. $X^\top X = \text{diag}(4, 20)$ and $X^\top y = (10, 12)^\top$, so $\hat{\beta} = (5/2, 3/5)^\top$. The fit is $(7, 19, 31, 43)^\top/10$ and the residual is $(3, 1, -11, 7)^\top/10$. Its entries sum to zero, and its dot product with t is $(-9 - 1 - 11 + 21)/10 = 0$. The SSE is $(9 + 1 + 121 + 49)/100 = 9/5$. With $n - p = 2$, $s^2 = 9/10$ and the estimated covariance is $s^2(X^\top X)^{-1} = \text{diag}(9/40, 9/200)$.

Exercise 11.8 (Gauss–Markov in covariance order). State and prove the covariance-order optimality of ordinary least squares among linear unbiased coefficient estimators, including the equality case.

Solution. For B with $BX = I$, put $D = B - L$, where $L = (X^\top X)^{-1}X^\top$. Then $DX = 0$, and $LD^\top = (X^\top X)^{-1}(DX)^\top = 0$. Consequently $\text{Cov}(BY) = \sigma^2 BB^\top = \sigma^2 LL^\top + \sigma^2 DD^\top$. For any c , the excess variance of $c^\top BY$ is $\sigma^2 \|D^\top c\|^2 \geq 0$, proving the matrix order. Equality of the complete covariance matrices gives $0 = \text{tr}(DD^\top) = \sum_{i,j} d_{ij}^2$, so $B = L$; the converse is immediate. The hypotheses are fixed full-column-rank X , zero mean errors, and covariance $\sigma^2 I$ with $\sigma^2 > 0$.

Exercise 11.9 (Minimum total variance as a matrix problem). For a given full-column-rank $X \in \mathbb{R}^{n \times p}$, find the $p \times n$ matrix G minimizing $\text{tr}(GG^\top)$ subject to $GX = I_p$.

Solution. Let $L = (X^\top X)^{-1}X^\top$. Every feasible G is $L + D$ with $DX = 0$. The cross products $LD^\top$ and $DL^\top$ vanish, so

$$\text{tr}(GG^\top) = \text{tr}(LL^\top) + \text{tr}(DD^\top) = \text{tr}((X^\top X)^{-1}) + \sum_{i,j} d_{ij}^2.$$

The second term is nonnegative and is zero exactly when $D = 0$. Thus the unique minimizer is $G = L$. This solves the stated scalar trace minimization; the preceding covariance-order theorem is stronger.

Exercise 11.10 (Centered sum-of-squares decomposition). When an intercept is present, prove $\text{SST} = \text{SSR} + \text{SSE}$, with all three sums centered or uncentered as defined in the notes.

Solution. Put $J = \mathbf{1}\mathbf{1}^\top/n$. The intercept gives $HJ = JH = J$. Then $H - J$ and $M = I - H$ are orthogonal projectors with $(H - J)M = 0$ and $(H - J) + M = I - J$. Hence $(I - J)y = (H - J)y + My$ is an orthogonal sum. Taking squared lengths gives

$$\|y - \bar{y}\mathbf{1}\|^2 = \|Hy - \bar{y}\mathbf{1}\|^2 + \|My\|^2.$$

These are exactly SST, SSR and SSE. The argument depends on the intercept; without it, $HJ = J$ need not hold.

Exercise 11.11 (Nested model spaces). For nested model spaces $\mathcal{S}_0 \subseteq \mathcal{S}_1$, prove that their residual sums of squares satisfy $\text{SSE}_0 \geq \text{SSE}_1$.

Solution. Let H_i be the respective orthogonal projectors. Since H_1 fixes $\mathcal{S}_0$, $H_1 H_0 = H_0$; transposition gives $H_0 H_1 = H_0$. Thus $H_1 - H_0$ is a symmetric idempotent matrix orthogonal to $I - H_1$. The residual decomposition $(I - H_0)y = (H_1 - H_0)y + (I - H_1)y$ gives $\text{SSE}_0 - \text{SSE}_1 = \|(H_1 - H_0)y\|^2 \geq 0$. Equality holds precisely when the added model directions receive no component of the observed response.

Exercise 11.12 (Partitioned regression). Let $X = (X_1 \ X_2)$ have full column rank and $M_2 = I - X_2(X_2^\top X_2)^{-1}X_2^\top$. Prove that the first coefficient block is $(X_1^\top M_2 X_1)^{-1}X_1^\top M_2 Y$, and that its residualized regression has the same residuals as the full regression.

Solution. If $M_2 X_1 v = 0$, then $X_1 v = X_2 u$ for some u . Full column rank of $(X_1 \ X_2)$ forces $v = u = 0$, so $X_1^\top M_2 X_1$ is positive definite. For fixed b_1 , the least-squares b_2 equals $(X_2^\top X_2)^{-1}X_2^\top (Y - X_1 b_1)$. Substitution makes the residual $M_2 Y - M_2 X_1 b_1$. The normal equations for this remaining full-column-rank problem are $X_1^\top M_2 X_1 b_1 = X_1^\top M_2 Y$, giving the formula. At its solution the displayed residual is exactly the full residual, because the eliminated b_2 was its unique optimal value for that b_1 .

Exercise 11.13 (Omitted-variable bias). Suppose $EY = X_1 \beta_1 + X_2 \beta_2$, but fit only the full-column-rank design X_1 . Find the expectation of the reduced coefficient estimator and the condition for it to be unbiased for β_1 .

Solution. The reduced estimator is $\tilde{\gamma} = (X_1^\top X_1)^{-1}X_1^\top Y$. Taking its expectation gives

$$E\tilde{\gamma} = \beta_1 + (X_1^\top X_1)^{-1}X_1^\top X_2 \beta_2.$$

For a specified β_2 , unbiasedness holds exactly when $X_1^\top X_2 \beta_2 = 0$. For all possible β_2 , it holds exactly when $X_1^\top X_2 = 0$. A zero omitted coefficient vector is sufficient in the first statement; it is not necessary, since cancellations in the matrix-vector product are possible.

Exercise 11.14 (Prediction variance). Let $Y_0 = x_0^\top \beta + \varepsilon_0$ be a new observation with $E\varepsilon_0 = 0$, $\text{Var}(\varepsilon_0) = \sigma^2$, and $\text{Cov}(\varepsilon_0, \varepsilon) = 0$. For $\hat{Y}_0 = x_0^\top \hat{\beta}$, compare the variance of estimating the mean $x_0^\top \beta$ with that of predicting Y_0 .

Solution. The first error is $x_0^\top (\hat{\beta} - \beta)$, with variance $\sigma^2 x_0^\top (X^\top X)^{-1} x_0$. The prediction error subtracts ε_0 . Its cross-covariance with the coefficient error is zero by the assumed error cross-covariance, so its variance is $\sigma^2 [1 + x_0^\top (X^\top X)^{-1} x_0]$. It exceeds the mean-estimation variance by $\sigma^2 > 0$, the variability of the new observation about its own mean.

Exercise 11.15 (Adjusted explained variation). For two full-rank regressions sharing an intercept and differing by one added column, assume SST is positive and $n - p - 1 > 0$, where p is the reduced parameter count. Prove that R^2 cannot decrease. Define

$$\bar{R}^2 = 1 - \frac{\text{SSE}/(n - \text{parameter count})}{\text{SST}/(n - 1)}.$$

Show that $\bar{R}^2$ strictly increases exactly when the added coefficient's $|t|$ exceeds 1, where t uses the full-model residual variance and is defined when that variance is positive.

Solution. Nested-model geometry gives $\text{SSE}_F \leq \text{SSE}_R$. Since SST is the same positive number, $R^2 = 1 - \text{SSE}/\text{SST}$ cannot decrease. Let z be the new column residualized against the old design; full rank gives $z \neq 0$. Its fitted coefficient is $\hat{\gamma} = z^\top y / (z^\top z)$, and the SSE decrease is $\Delta = \hat{\gamma}^2 z^\top z$. Put $d = n - p - 1$ and $s_F^2 = \text{SSE}_F/d > 0$. Then $t^2 = \hat{\gamma}^2 / (s_F^2 / (z^\top z)) = \Delta / s_F^2$. The adjusted value increases exactly when $\text{SSE}_F/d < \text{SSE}_R/(d + 1)$, or $\text{SSE}_F < d\Delta$, equivalently $t^2 > 1$. This is an algebraic identity; no distributional interpretation of t is needed. If the full SSE is zero, t in this form is undefined.

Weighted and constrained estimation

Exercise 11.16 (Generalized least squares). For full-column-rank X and positive definite Ω , find the unique minimizer of $(y - Xb)^\top \Omega^{-1}(y - Xb)$.

Solution. Set $Z = \Omega^{-1/2}X$, $v = \Omega^{-1/2}y$. Since the whitening matrix is invertible, Z has full column rank. The objective becomes $\|v - Zb\|^2$, whose unique minimizer is

$$b_0 = (Z^\top Z)^{-1}Z^\top v = (X^\top \Omega^{-1}X)^{-1}X^\top \Omega^{-1}y.$$

Equivalently, for every displacement d , the objective increase from b_0 to $b_0 + d$ is $d^\top X^\top \Omega^{-1}Xd$, strictly positive for $d \neq 0$. This also verifies global uniqueness.

Exercise 11.17 (Actual OLS covariance and optimal GLS). Under $\text{Cov}(\varepsilon) = \sigma^2\Omega \succ 0$, prove that OLS is still unbiased, derive its actual covariance, and prove that GLS is best linear unbiased in covariance order.

Solution. With $K = X^\top X$, $L = K^{-1}X^\top$ still satisfies $LX = I$. Thus $E(LY) = \beta$ and $\text{Cov}(LY) = \sigma^2 K^{-1}X^\top \Omega X K^{-1}$. Let $L_\Omega = (X^\top \Omega^{-1}X)^{-1}X^\top \Omega^{-1}$. It satisfies $L_\Omega X = I$ and has covariance $\sigma^2(X^\top \Omega^{-1}X)^{-1}$. For any unbiased BY , put $D = B - L_\Omega$; then $DX = 0$ and $L_\Omega \Omega D^\top = 0$. Expansion gives $\text{Cov}(BY) - \text{Cov}(L_\Omega Y) = \sigma^2 D \Omega D^\top \succeq 0$. The sandwich formula is the correct OLS covariance under this model; the spherical formula generally fails.

Exercise 11.18 (Minimum total variance for a general target). Let $X \in \mathbb{R}^{n \times p}$ have full column rank, $\Omega \succ 0$, and $W \in \mathbb{R}^{m \times p}$ be fixed. Minimize $\text{tr}(G\Omega G^\top)$ over $G \in \mathbb{R}^{m \times n}$ subject to $GX = W$.

Solution. Let $K = X^\top \Omega^{-1}X$ and $G_0 = WK^{-1}X^\top \Omega^{-1}$. It is feasible because $G_0 X = W$. Every feasible G is $G_0 + D$ with $DX = 0$, and $G_0 \Omega D^\top = WK^{-1}(DX)^\top = 0$. Therefore $\text{tr}(G\Omega G^\top) = \text{tr}(WK^{-1}W^\top) + \text{tr}(D\Omega D^\top)$. The last term is $\|D\Omega^{1/2}\|_F^2$, zero only when $D = 0$. Thus G_0 is the unique minimizer, including when W has dependent or zero rows.

Exercise 11.19 (A matrix Cauchy–Schwarz inequality). Suppose $A, B \in \mathbb{R}^{n \times p}$ and $A^\top B = I_p$. Prove $A^\top A \succeq (B^\top B)^{-1}$ and $B^\top B \succeq (A^\top A)^{-1}$, and characterize equality.

Solution. $Bx = 0$ implies $x = A^\top Bx = 0$, so $B^\top B$ is positive definite. Let $P = B(B^\top B)^{-1}B^\top$. Since $B^\top A = I$,

$$A^\top A - (B^\top B)^{-1} = A^\top(I - P)A = [(I - P)A]^\top [(I - P)A] \succeq 0.$$

Equality holds exactly when $(I - P)A = 0$, or $A = PA = B(B^\top B)^{-1}$. For the second inequality, $B^\top A = I$ also gives full column rank of A . With $Q = A(A^\top A)^{-1}A^\top$, the difference is $B^\top(I - Q)B \succeq 0$, zero exactly when $B = A(A^\top A)^{-1}$. The equality conditions are equivalent by substitution into either Gram matrix.

Exercise 11.20 (Two proofs of the OLS–GLS inequality). Prove, for full-column-rank X and $V \succ 0$,

$$(X^\top X)^{-1}X^\top V X (X^\top X)^{-1} \succeq (X^\top V^{-1}X)^{-1},$$

first using the preceding matrix Cauchy–Schwarz inequality, then using the projector onto $\text{col}(V^{-1/2}X)$. Explain its statistical meaning.

Solution. Set $A = V^{1/2}X(X^\top X)^{-1}$ and $B = V^{-1/2}X$. Then $A^\top B = I$, and the preceding inequality gives exactly the displayed formula. For a second proof put $P = V^{-1/2}X(X^\top V^{-1}X)^{-1}X^\top V^{-1/2}$. It is an orthogonal projector, so

$$V^{1/2}(I - P)V^{1/2} = V - X(X^\top V^{-1}X)^{-1}X^\top \succeq 0.$$

Multiplying on both sides by the transpose pair $(X^\top X)^{-1}X^\top$ and $X(X^\top X)^{-1}$ gives the claim. When $\text{Cov}(Y) = V$, the two sides are the actual covariances of OLS and GLS. Thus GLS never has a larger variance for any coefficient contrast. Equality can occur; OLS need not be strictly worse.

Exercise 11.21 (* When OLS and GLS agree). Let $K = X^\top X$, $J = X^\top V^{-1}X$, $M = I - XK^{-1}X^\top$, with X of full column rank and $V \succ 0$. Prove the equivalence of

$$X^\top V^{-1}M = 0, \quad J^{-1}X^\top V^{-1} = K^{-1}X^\top, \quad J^{-1} = K^{-1}X^\top V X K^{-1}.$$

Solution. Expanding the first equality gives $X^\top V^{-1} = JK^{-1}X^\top$; multiplying by J^{-1} proves its equivalence with the second. Let $L = K^{-1}X^\top$, $L_V = J^{-1}X^\top V^{-1}$. If $L = L_V$, their covariance matrices $LVL^\top$ and $L_V V L_V^\top = J^{-1}$ agree, giving the third equality. Conversely set $D = L - L_V$. Since $DX = 0$, both cross terms in $(L_V + D)V(L_V + D)^\top$ vanish. Thus

$$K^{-1}X^\top V X K^{-1} - J^{-1} = D V D^\top.$$

The third equality makes the right side zero. Taking its trace gives $\|DV^{1/2}\|_F^2 = 0$; invertibility of $V^{1/2}$ implies $D = 0$, which is the second equality. This proves all directions, including the nontrivial implication from equal covariance to equal estimator matrices.

Exercise 11.22 (A stronger covariance bound). For full-column-rank X and $V \succ 0$, prove $X(X^\top V^{-1}X)^{-1}X^\top \preceq V$. Deduce $(X^\top V^{-1}X)^{-1} \preceq X^\top V X$ when $X^\top X = I$.

Solution. The projector P onto $\text{col}(V^{-1/2}X)$ satisfies $0 \preceq P \preceq I$. Multiplication by $V^{1/2}$ on both sides gives $X(X^\top V^{-1}X)^{-1}X^\top \preceq V$. If $X^\top X = I$, multiply this inequality on the left by $X^\top$ and on the right by X . The left side reduces to the requested inverse. Congruence preserves the semidefinite order because its quadratic forms are the original quadratic forms evaluated on transformed vectors.

Exercise 11.23 (Weighted averaging). For $X = (1, 1)^\top$, $y = (1, 3)^\top$, and $\Omega = \text{diag}(1, 4)$, compute OLS and GLS, their variances under covariance $\sigma^2\Omega$, and the GLS fitted matrix and residual. Verify weighted orthogonality.

Solution. OLS is $(1 + 3)/2 = 2$ and its variance is $\sigma^2(1/2, 1/2)\Omega(1/2, 1/2)^\top = 5\sigma^2/4$. For GLS, $X^\top \Omega^{-1}X = 5/4$ and $X^\top \Omega^{-1}y = 7/4$, so the estimate is $7/5$ and its variance is $4\sigma^2/5$. The fitted matrix is $H_\Omega = \begin{pmatrix} 4/5 & 1/5 \\ 4/5 & 1/5 \end{pmatrix}$, the fit is $(7/5, 7/5)^\top$, and the residual is $(-2/5, 8/5)^\top$. The weighted residual inner product with X is $-2/5 + (1/4)(8/5) = 0$. Although H_Ω is not symmetric, its square equals itself because its two row entries sum to one.

Exercise 11.24 (Constrained weighted least squares). For full-column-rank X , $\Omega \succ 0$, and full-row-rank $R \in \mathbb{R}^{q \times p}$, minimize $(y - Xb)^\top \Omega^{-1}(y - Xb)$ subject to $Rb = r$. Find the exact increase above the unconstrained minimum.

Solution. Put $K = X^\top \Omega^{-1}X$, $b_0 = K^{-1}X^\top \Omega^{-1}y$, $C = RK^{-1}R^\top$, and $d = Rb_0 - r$. Both K, C are positive definite. The candidate is $b_R = b_0 - K^{-1}R^\top C^{-1}d$, and $Rb_R = Rb_0 - CC^{-1}d = r$. Every feasible b has form $b_R + v$ with $Rv = 0$. Weighted normal equations give $Q(b) = Q(b_0) + (b - b_0)^\top K(b - b_0)$. Since $v^\top K(b_R - b_0) = -v^\top R^\top C^{-1}d = 0$,

$$Q(b) = Q(b_0) + d^\top C^{-1}d + v^\top K v.$$

The last term is strictly positive for $v \neq 0$, proving uniqueness, and the middle term is the exact increase.

Exercise 11.25 (Constrained OLS by multipliers). Derive the constrained ordinary least-squares formula under $Rb = r$ using Lagrange multipliers, where X has full column rank and R has full row rank. Prove that its SSE is at least the unconstrained SSE.

Solution. Use $\mathcal{L}(b, \lambda) = \frac{1}{2} \|y - Xb\|^2 + \lambda^\top (Rb - r)$. Differentiation gives $Kb - X^\top y + R^\top \lambda = 0$ and $Rb = r$, where $K = X^\top X$. Thus $b = b_0 - K^{-1}R^\top \lambda$ and $RK^{-1}R^\top \lambda = Rb_0 - r$. Since the coefficient matrix is positive definite, this determines λ uniquely and gives $b_R = b_0 - K^{-1}R^\top (RK^{-1}R^\top)^{-1}(Rb_0 - r)$. To verify that the stationary point is a global constrained minimum, write any other feasible $b = b_R + v$ with $Rv = 0$ and use the orthogonal completion in the preceding solution. The SSE increase is the nonnegative quadratic form $(Rb_0 - r)^\top (RK^{-1}R^\top)^{-1}(Rb_0 - r)$.

Exercise 11.26 (A numerical restriction). For the data in Exercise 11.7, impose the restriction that the slope is zero. Compute the constrained coefficients and SSE, and verify the increase formula.

Solution. The fitted constant is the mean $5/2$. Thus the constrained coefficient column is $(5/2, 0)^\top$ and its residual is $(-3/2, -1/2, -1/2, 5/2)^\top$, with squared length 9. Here $R = (0 \ 1)$, $K^{-1} = \text{diag}(1/4, 1/20)$, $RK^{-1}R^\top = 1/20$ and $Rb_0 = 3/5$. The increase formula gives $(3/5)^2/(1/20) = 36/5$; adding the unconstrained SSE $9/5$ gives 9, as computed directly.

Exercise 11.27 (* Rank-deficient estimation of the mean). Let $X \in \mathbb{R}^{n \times p}$ have arbitrary rank and $\Omega \succ 0$. Minimize $\text{tr}(G\Omega G^\top)$ over $n \times n$ matrices satisfying $GX = X$.

Solution. Put $Z = \Omega^{-1/2}X$, $P = ZZ^\top$ and $G_0 = \Omega^{1/2}P\Omega^{-1/2}$. Then $G_0X = \Omega^{1/2}PZ = X$, so it is feasible. Every feasible G is $G_0 + D$ with $DX = 0$. Let $E = D\Omega^{1/2}$. Then $EZ = 0$, which implies $EP = EZZ^\top = 0$. The Frobenius cross term between $G_0\Omega^{1/2} = \Omega^{1/2}P$ and E is $\text{tr}(\Omega^{1/2}PE^\top) = 0$. Therefore

$$\text{tr}(G\Omega G^\top) = \text{tr}(G_0\Omega G_0^\top) + \|E\|_F^2,$$

uniquely minimized by $D = 0$. Using the SVD identity $Z(Z^\top Z)^\dagger Z^\top = P$, the answer also equals $G_0 = X(X^\top \Omega^{-1}X)^\dagger X^\top \Omega^{-1}$. If $X = 0$, the constraint is vacuous and this formula gives $G_0 = 0$, the correct unique minimizer.

Exercise 11.28 (* A residual quadratic minimization). Let X have rank r . Minimize $n^{-1} \text{tr}(V^2)$ over symmetric positive semidefinite $V \in \mathbb{R}^{n \times n}$ satisfying $VX = 0$ and $\text{tr} V = 1$. Treat the case $r = n$ as well as $r < n$.

Solution. If $r = n$, X spans $\mathbb{R}^n$, so $VX = 0$ forces $V = 0$, contradicting $\text{tr} V = 1$. Thus the feasible set is empty. If $r < n$, let Q have orthonormal columns spanning $\ker X^\top$, of dimension $d = n - r$. Symmetry and $VX = 0$ imply $V = QWQ^\top$ with $W = Q^\top VQ \succeq 0$. Then $\text{tr} W = 1$ and $\text{tr}(V^2) = \text{tr}(W^2)$. If $\lambda_1, \dots, \lambda_d \geq 0$ are the eigenvalues of W , Cauchy–Schwarz gives $1 = (\sum_i \lambda_i)^2 \leq d \sum_i \lambda_i^2$. Equality holds exactly when all $\lambda_i = 1/d$. Hence the unique minimizer is $V = QQ^\top/d = (I - XX^\dagger)/(n - r)$ and the minimum is $1/[n(n - r)]$.

Exercise 11.29 (Complementary spaces and a determinant identity). Suppose $X \in \mathbb{R}^{n \times p}$ and $A \in \mathbb{R}^{n \times (n-p)}$ both have full column rank, $A^\top X = 0$, and $\Omega \succ 0$. Prove

$$\det(A^\top \Omega A) \det(X^\top X) = \det(A^\top A) \det(X^\top \Omega^{-1} X) \det \Omega.$$

Solution. Choose orthonormal bases U, V of the two complementary column spaces, and write $X = UR$, $A = VS$ with square invertible R, S . All factors $\det(R)^2$ and $\det(S)^2$ cancel, reducing the assertion to $\det(V^\top \Omega V) = \det(U^\top \Omega^{-1} U) \det \Omega$. In the orthogonal basis $(U \ V)$ write $\Omega = \begin{pmatrix} B & C \\ C^\top & D \end{pmatrix}$. Here $D \succ 0$. Block elimination gives $\det \Omega = \det D \det(B - CD^{-1}C^\top)$. Solving the block system for the first block column of the inverse gives $U^\top \Omega^{-1} U = (B - CD^{-1}C^\top)^{-1}$. Taking determinants proves the reduced identity and hence the original. The endpoint cases use the convention that an empty determinant is 1.

Random-variable spaces and sensitivity: additional reading

Exercise 11.30 (* Square-integrable random variables). On a fixed probability space, show that the real random variables with finite second moment form a vector space, and that the finite linear combinations of any such $Z_1, \dots, Z_k$ form a subspace. Identify random variables equal with probability one, and explain the zero vector in this convention.

Solution. The pointwise inequality $(U + V)^2 \leq 2U^2 + 2V^2$ gives $E(U + V)^2 \leq 2E U^2 + 2E V^2 < \infty$. Also $E(cU)^2 = c^2 E U^2 < \infty$. Thus addition and scaling preserve finite second moments. The ordinary pointwise real-number identities give all vector-space axioms. If representatives are changed on sets of probability zero, sums and scalar multiples change only on unions of such sets, again of probability zero, so the operations on classes are well-defined. The zero class consists of variables zero with probability one. Finally, linear combinations of the finite list remain such combinations after adding or scaling their coefficients, so they form a subspace. Multiple coefficient representations do not make the random variable itself ambiguous.

For the following exercises, the differential is the linear part of a change: $df(x)[h] = \left. \frac{d}{dt} f(x + th) \right|_{t=0}$. For scalar f , its derivative is the row $Df(x)$ satisfying $df(x)[h] = Df(x)h$. Matrix derivatives may be specified as linear maps of a perturbation matrix. Product and transpose rules follow entry by entry from ordinary differentiation. Differentiating $A^{-1}A = I$ gives $d(A^{-1}) = -A^{-1}(dA)A^{-1}$ wherever A is invertible.

Exercise 11.31 (Linear, quadratic, and bilinear differentiation). For constant a, A , find the differential and derivative of $a^\top x$, $x^\top Ax$, and $x_1^\top Ax_2$ with respect to the stacked variable $(x_1^\top, x_2^\top)^\top$.

Solution. The first differential is $a^\top dx$, so the derivative is $a^\top$. For the second, $d(x^\top Ax) = (dx)^\top Ax + x^\top A dx = x^\top (A + A^\top) dx$, so its derivative is $x^\top (A + A^\top)$. For the third, $d(x_1^\top Ax_2) = x_2^\top A^\top dx_1 + x_1^\top A dx_2$, so the derivative is the block row $(x_2^\top A^\top \quad x_1^\top A)$. These expressions include every first-order term; the product of two perturbations contributes only a second-order term.

Exercise 11.32 (Least squares by differentiation). Derive the ordinary least-squares formula by differentiating $Q(b) = \|y - Xb\|^2$, and verify the second-order condition.

Solution. Expansion gives $Q(b) = y^\top y - 2b^\top X^\top y + b^\top K b$ with $K = X^\top X \succ 0$. Thus $dQ = 2(Kb - X^\top y)^\top db$, giving the stationary point $b_0 = K^{-1}X^\top y$. The Hessian is $2K \succ 0$. For global rather than merely local minimality, $Q(b_0 + d) - Q(b_0) = d^\top K d > 0$ for every $d \neq 0$.

Exercise 11.33 (Sensitivity of the coefficient estimator). With y fixed and X of full column rank, obtain the differential of $b = (X^\top X)^{-1}X^\top y$ as a linear map of dX . Then give its derivative when only column x_j of X varies.

Solution. Let $K = X^\top X$ and $e = y - Xb$. Differentiate the identity $Kb = X^\top y$:

$$K db + (dX)^\top Xb + X^\top (dX)b = (dX)^\top y.$$

Solving gives $db = K^{-1}(dX)^\top e - K^{-1}X^\top (dX)b$. This is the derivative as a linear map from matrices to columns. If only column j changes, $dX = (dx_j)d_j^\top$, with d_j the j th coordinate column of $\mathbb{R}^p$. Hence

$$db = K^{-1}(d_j e^\top - b_j X^\top) dx_j,$$

and the displayed $p \times n$ matrix is the derivative with respect to $x_j^\top$. Full column rank persists in a sufficiently small neighborhood because $\det(X^\top X) > 0$ and is continuous.

Exercise 11.34 (Sensitivity of the residual). For the same problem, obtain the differential of $e = y - Xb$ as a linear map of dX , and its derivative when only x_j varies.

Solution. Using $de = -(dX)b - Xdb$ and the previous coefficient differential gives

$$de = -M(dX)b - XK^{-1}(dX)^\top e.$$

For $dX = (dx_j)d_j^\top$, collect the terms to get $de = -(b_jM + XK^{-1}d_je^\top)dx_j$. Thus the derivative is $-(b_jM + XK^{-1}d_je^\top)$. The first term changes the observed residualized fitted column; the second accounts for the adjustment of the estimated coefficients.

Exercise 11.35 (Sensitivity of generalized least squares). Let the positive definite matrix $\Omega(t)$ vary differentiably with a real parameter, while X, y stay fixed. Find the derivative of $b(t) = (X^\top\Omega^{-1}X)^{-1}X^\top\Omega^{-1}y$.

Solution. Put $K = X^\top\Omega^{-1}X$ and $e = y - Xb$. Differentiating the weighted normal equations $X^\top\Omega^{-1}(y - Xb) = 0$ gives $K\dot{b} = X^\top\frac{d\Omega^{-1}}{dt}e$. The inverse rule gives $\frac{d\Omega^{-1}}{dt} = -\Omega^{-1}\dot{\Omega}\Omega^{-1}$. Therefore

$$\dot{b} = -K^{-1}X^\top\Omega^{-1}\dot{\Omega}\Omega^{-1}e.$$

This answers the scalar-parameter sensitivity question directly; no vectorization or Kronecker notation is needed.

Exercise 11.36 (Sensitivity of the residual projector). For full-column-rank X and $M = I - X(X^\top X)^{-1}X^\top$, obtain the differential of M as a linear map of dX . Specialize it to perturbing only column j .

Solution. With $K = X^\top X$, product and inverse differentiation give

$$\begin{aligned} dM &= -(dX)K^{-1}X^\top - XK^{-1}(dX)^\top \\ &\quad + XK^{-1}[(dX)^\top X + X^\top(dX)]K^{-1}X^\top. \end{aligned}$$

Collect the first and fourth terms, and the second and third terms:

$$dM = -M(dX)K^{-1}X^\top - XK^{-1}(dX)^\top M.$$

If $dX = (dx_j)d_j^\top$, put $v = XK^{-1}d_j$ to obtain $dM = -M(dx_j)v^\top - v(dx_j)^\top M$. This explicitly gives the linear derivative for either type of perturbation and shows that dM is symmetric.